

701: Enterprise Solutions Architecture

Lloyd Evans
Partner Systems Engineering

PalmSource
Conference 2000

Overview

- Architecture Options 1 2 3
  - *What* to use (handheld applications)
  - *How* to access (connectivity)
  - *Where* to connect (servers)
- Example Applications
- Resources

Architecture Options 1 2 3

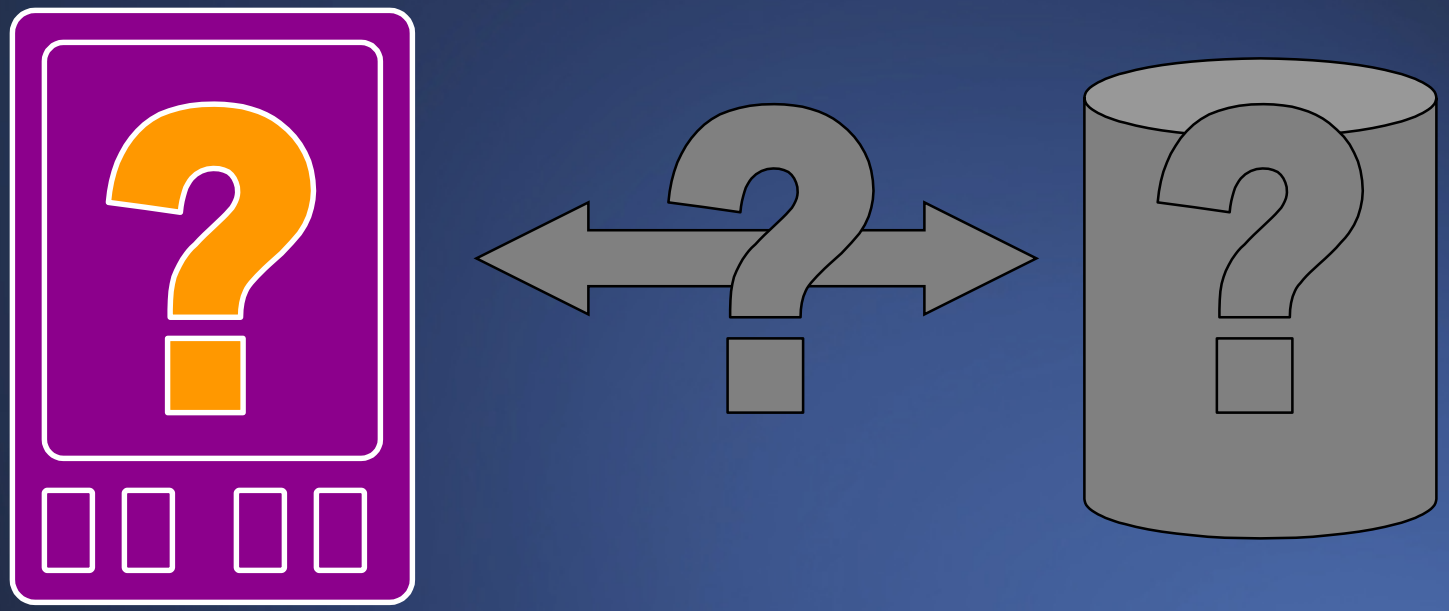


1 What?

2 How?

3 Where?

1 Applications



1 What?

Palm OS® Applications

- Rich user interface
- Full use of handheld (storage, IR, alarms, etc)
- Developed with PalmOS tools
  - CodeWarrior for “C” developers
  - SatForms, Pendragon, MobileBuilder for forms
- Examples
  - MultiMail, DateBk4
  - Heavy user interaction

Browser Applications

- No Palm OS development needed
- Use existing 3rd party browsers
  - HTML (Palmscape, Handweb, Web Clipping)
  - WAP (KBrowser, WAPman)
- Limited user interface
- Little or no device interaction
- Examples
  - E*Trade, AA.com
  - Quick access to remote data

2 Connectivity



2 How?

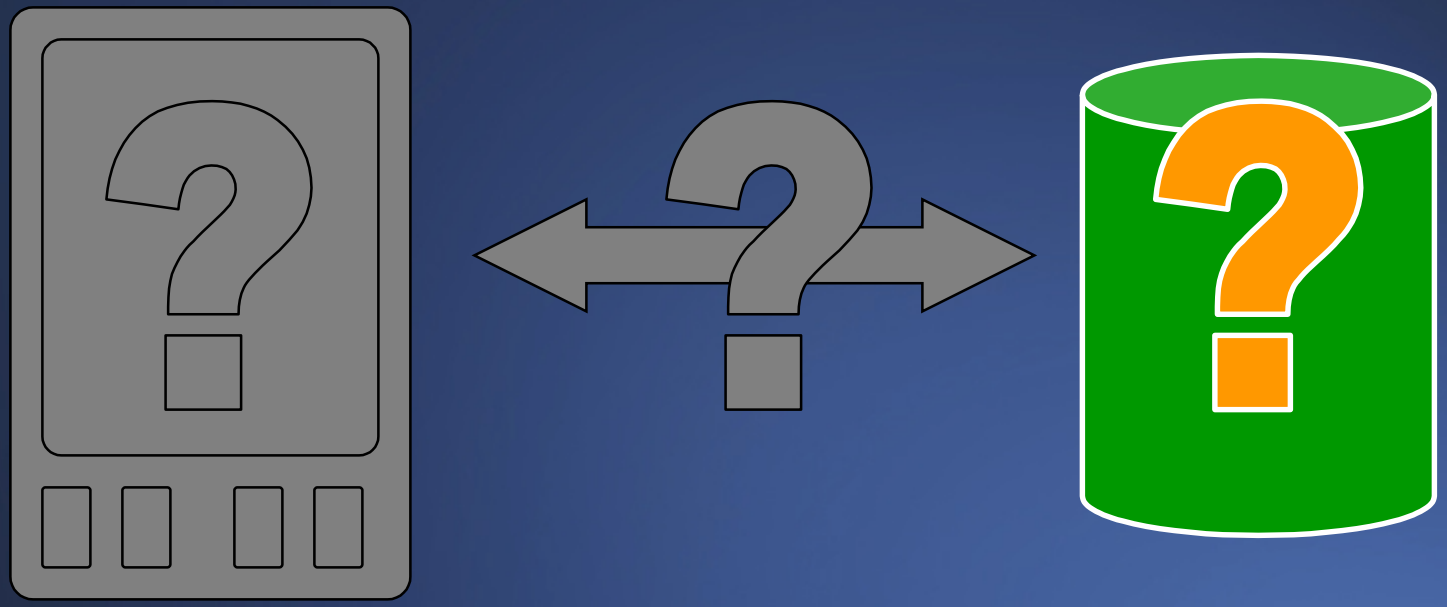
Batch Synchronization

- Typically used wire-line
  - Desktop cradle
  - Ethernet cradle
- Fast speed, continuous connection
- Large updates of data
  - Phone books, price lists, product catalogs
- Non- or near-realtime

Real-time Access

- Typically used wirelessly
  - Wireless device (Palm VII)
  - Wireless modem (Omnisky)
- Low speed, intermittent connection
- Transfer of small amounts of data
  - Order status, shipment status, price change

3 Server Systems

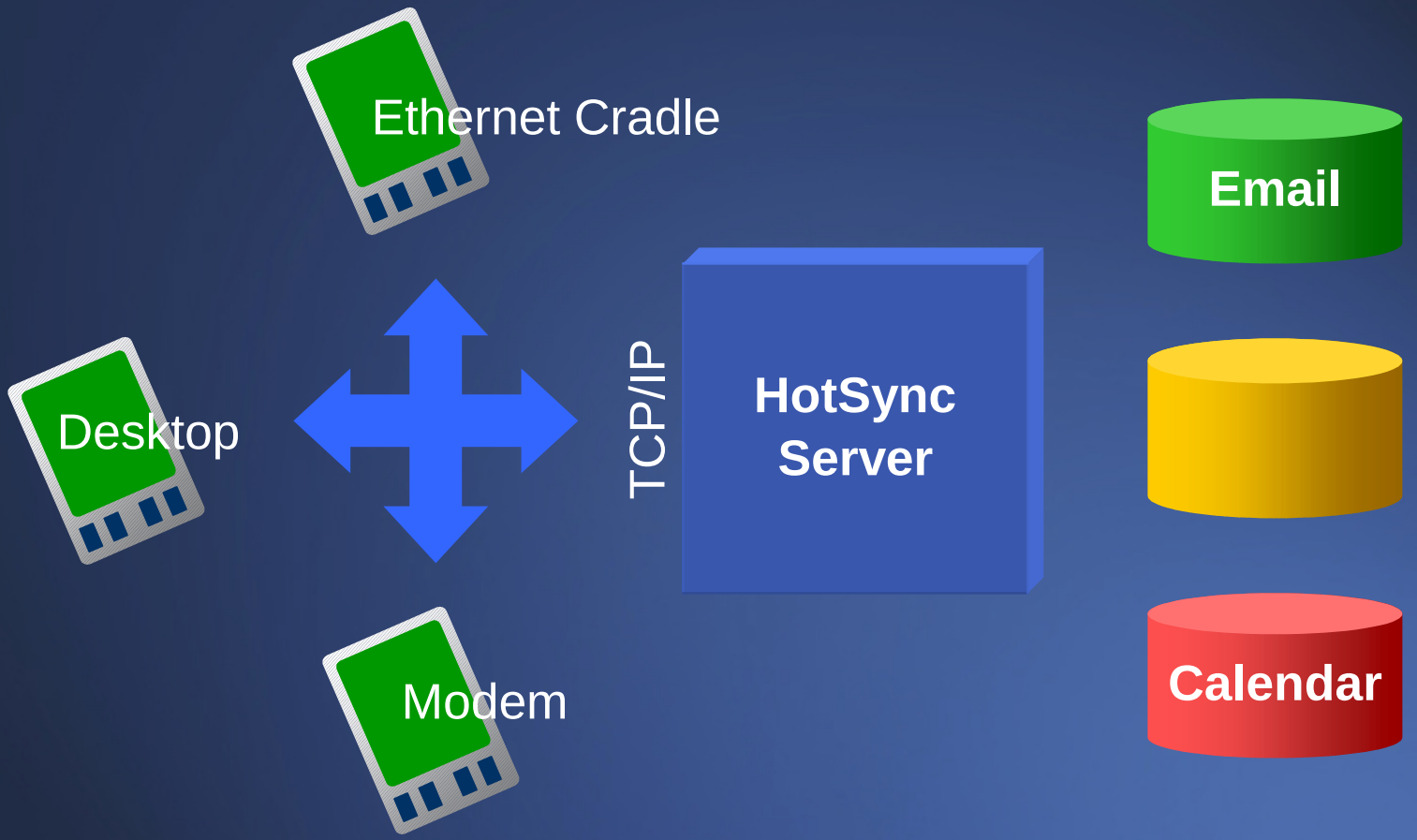


3 Where?

Synchronization Servers

- Multiple simultaneous users with multiple access options
- Central administration
- “Plug in” model to support multiple applications
- Examples
  - HotSync Server
  - IntelliSync Anywhere
  - IBM MobileConnect

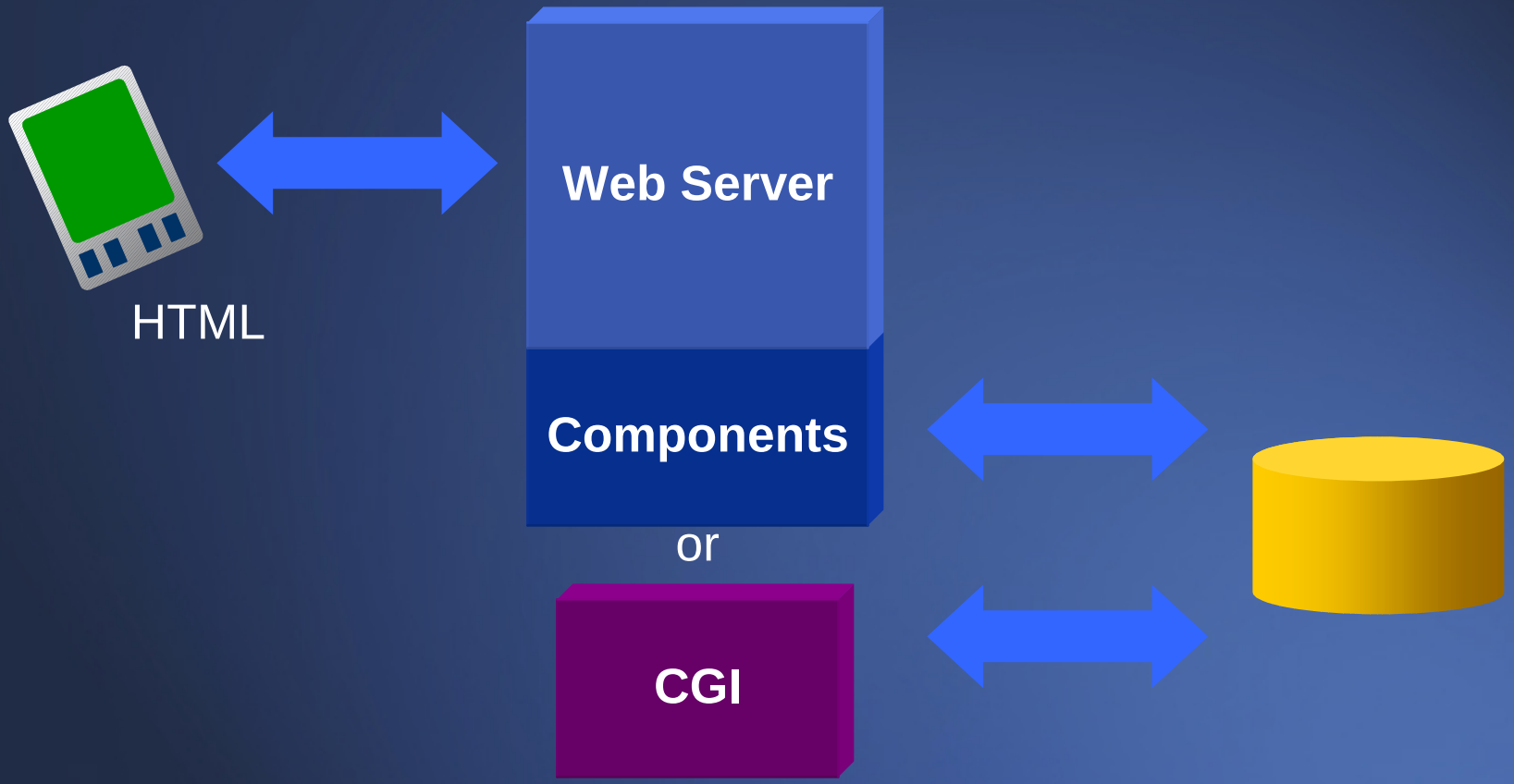
Sync Server Diagram



Web Servers

- Extend existing web-based applications to mobile users
- HTML, WML only
- Repurpose content for small screens
- Requires multiple “sub” sites, or intelligent scripts

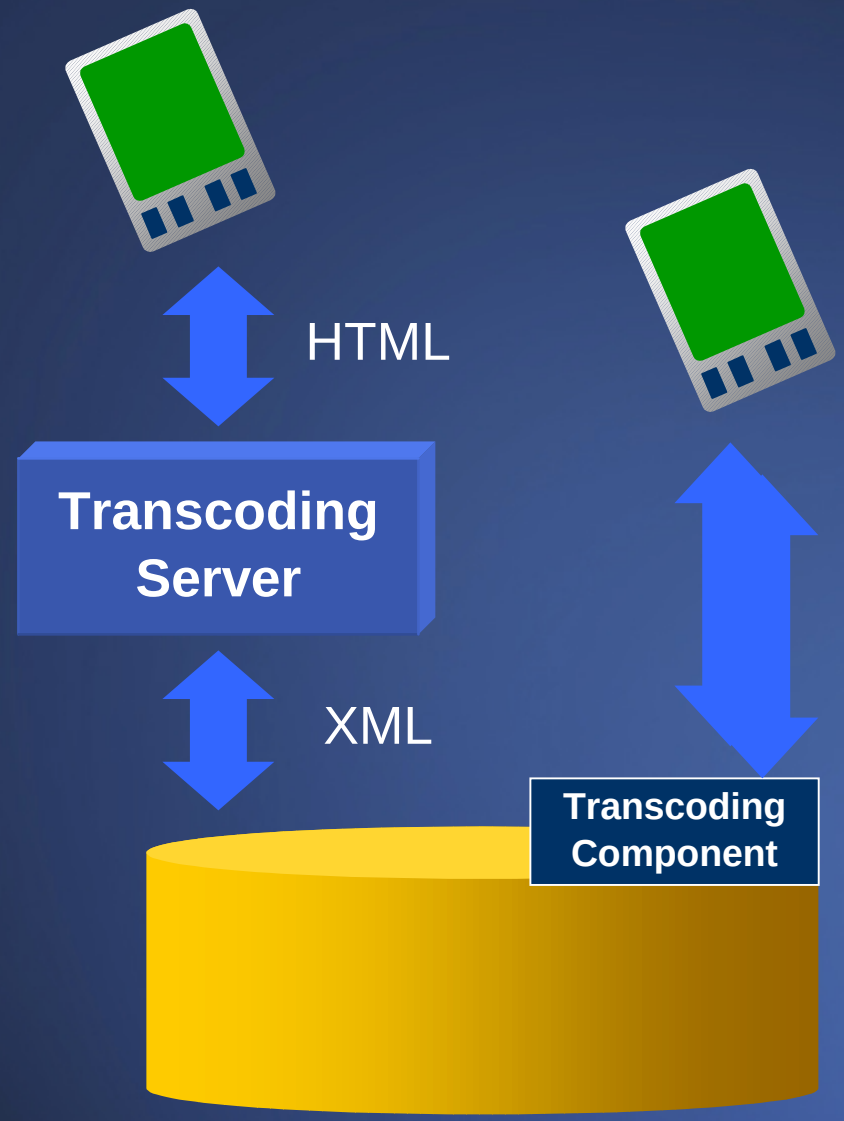
Web Server Diagram



Transcoding

- One application supporting various rendering options for various clients
- Transformation of data according to client device capabilities, or
  - Typically XML based
- Translation of existing data (screen scraping)
  - HTML conversion, reformatting
- Examples:
  - IBM Transcoding Publisher
  - Oracle Applications Server Wireless Edition

Transcoding Diagram



Example 1: Siebel Systems, Inc

- Mobile Field Sales
- Native PalmOS application
  - Full user interface
  - Local data storage
- Synchronization for batch updates
  - Desktop synchronization
  - Direct server synchronization

Example 2: SAP

- Sales Order Status
- Browser based application (Web Clipping)
- Simple user interface
- Real-time access to changing information
- Server-side templates used to transform SAP into small-screen HTML

Example 3: AvantGo

- Browser-based application
  - Re-use of existing HTML applications & content
  - Limited user interface
  - No advanced device interaction
- Synchronization
  - Mostly large downloads of data from servers
  - Synchronization from desktop or LAN
  - Wireless updates available
- HTTP Servers
  - Reusing existing system